

Uniform Resource Identifier

A **Uniform Resource Identifier** (**URI**), formerly **Universal Resource Identifier**, is a unique sequence of characters that identifies an abstract or physical resource,^{[1]:1} such as resources on a webpage, email address, phone number,^{[1]:7} books, real-world objects such as people and places, and concepts. In particular, the resource need not be retrievable via the Internet, or any computer network.^{[1]:5}

URIs which provide a means of locating and retrieving information resources on a network (like the Internet, an Intranet or a computer file system) are Uniform Resource Locators (URLs). URLs are therefore a subset of URIs.^{[1]:7} Other URIs provide only a unique name, without a means of locating or retrieving the resource or information about it; these are Uniform Resource Names (URNs). The web technologies that use URIs are not limited to web browsers.

Uniform Resource Identifier

Abbreviation URI

Native name RFC 3986 (<https://www.rfc-editor.org/rfc/rfc3986>)^[1]

Status Active

Organization Internet Engineering Task Force

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Domain [World Wide Web](#)

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History

Conception

URIs and URLs have a shared history. In 1990, [Tim Berners-Lee's](#) proposals for hypertext implicitly introduced the idea of a URL as a short string representing a resource that is the target of a hyperlink.^[2] At the time, people referred to it as a "hypertext name"^[3] or "document name".

Over the next three and a half years, as the World Wide Web's core technologies of HTML, HTTP, and web browsers developed, a need to distinguish a string that provided an address for a resource from a string that merely named a resource emerged. Although not yet formally defined, the term *Uniform Resource Locator* came to represent the former, and the more contentious *Uniform Resource Name* came to represent the latter. In July 1992 Berners-Lee's report on the Internet Engineering Task Force (IETF) "UDI (Universal Document Identifiers) BOF" mentions URLs (as Uniform Resource Locators), URNs (originally, as Unique Resource Numbers), and the need to charter a new working group.^[4] In November 1992 the IETF "URI Working Group" met for the first time.^[5]

During the debate over defining URLs and URNs, it became evident that the concepts embodied by

the two terms were merely aspects of the fundamental, overarching, notion of resource *identification*. In June 1994, the IETF published [RFC 1630](https://www.rfc-editor.org/rfc/rfc1630) (<https://www.rfc-editor.org/rfc/rfc1630>), Berners-Lee's first *Request for Comments* that acknowledged the existence of URLs and URNs. Most importantly, it defined a formal syntax for *Universal Resource Identifiers* (i.e. URL-like strings whose precise syntaxes and semantics depended on their schemes). It also attempted to summarize the syntaxes of URL schemes in use at the time. It acknowledged – *but did not standardize*—the existence of relative URLs and fragment identifiers.^[6]

Refinement

In December 1994, [RFC 1738](https://www.rfc-editor.org/rfc/rfc1738) (<https://www.rfc-editor.org/rfc/rfc1738>)^[7] formally defined relative and absolute URLs, refined the general URL syntax, defined how to resolve relative URLs to absolute form, and better enumerated the URL schemes then in use. The agreed definition and syntax of URNs had to wait until the publication of IETF [RFC 2141](https://www.rfc-editor.org/rfc/rfc2141) (<https://www.rfc-editor.org/rfc/rfc2141>)^[8] in May 1997.

The publication of IETF [RFC 2396](https://www.rfc-editor.org/rfc/rfc2396) (<https://www.rfc-editor.org/rfc/rfc2396>)^[9] in August 1998 saw the URI syntax become a separate specification^[9] and most of the parts of RFCs 1630 and 1738 relating to URIs and URLs in general were revised and expanded by the IETF. The new RFC changed the meaning of *U* in *URI* from "Universal" to "Uniform."

In December 1999, [RFC 2732](https://www.rfc-editor.org/rfc/rfc2732) (<https://www.rfc-editor.org/rfc/rfc2732>)^[10] provided a minor update to [RFC 2396](https://www.rfc-editor.org/rfc/rfc2396), allowing URIs to accommodate [IPv6](#) addresses. A number of shortcomings discovered in the two specifications led to a community effort, coordinated by [RFC 2396](#) co-author Roy Fielding, that culminated in the publication of IETF [RFC 3986](https://www.rfc-editor.org/rfc/rfc3986) (<https://www.rfc-editor.org/rfc/rfc3986>)^[11] in January 2005. While obsoleting the prior standard, it did not render the details of existing URL schemes obsolete; [RFC 1738](#) continues to govern such schemes except where otherwise superseded. IETF [RFC 2616](https://www.rfc-editor.org/rfc/rfc2616) (<https://www.rfc-editor.org/rfc/rfc2616>)^[11] for example, refines the `http` scheme. Simultaneously, the IETF published the content of [RFC 3986](#) as the full standard STD 66, reflecting the establishment of the URI generic syntax as an official Internet protocol.

In 2001, the [World Wide Web Consortium's](#) (W3C) Technical Architecture Group (TAG) published a guide to [best practices](#) and canonical URIs for publishing multiple versions of a given resource.^[12] For example, content might differ by language or by size to adjust for capacity or settings of the device used to access that content.

In August 2002, IETF [RFC 3305](https://www.rfc-editor.org/rfc/rfc3305) (<https://www.rfc-editor.org/rfc/rfc3305>)^[13] pointed out that the term "URL" had, despite widespread public use, faded into near obsolescence, and serves only as a reminder that some URIs act as addresses by having schemes implying network accessibility, regardless of any such actual use. As URI-based standards such as [Resource Description Framework](#) make evident, resource identification need not suggest the retrieval of resource representations over the Internet, nor need they imply network-based resources at all.

The [Semantic Web](#) uses the HTTP URI scheme to identify both documents and concepts for practical uses, a distinction which has caused confusion as to how to distinguish the two. The TAG

published an e-mail in 2005 with a solution of the problem, which became known as the *httpRange-14 resolution*.^[14] The W3C subsequently published an Interest Group Note titled "Cool URIs for the Semantic Web", which explained the use of content negotiation and the HTTP 303 response code for redirections in more detail.^[15]

Design

URLs and URNs

A Uniform Resource Name (URN) is a URI that identifies a resource by name in a particular namespace. A URN may be used to talk about a resource without implying its location or how to access it. For example, in the International Standard Book Number (ISBN) system, *ISBN 0-486-27557-4* identifies a specific edition of the William Shakespeare play *Romeo and Juliet*. The URN for that edition would be *urn:isbn:0-486-27557-4*. However, it gives no information as to where to find a copy of that book.

A Uniform Resource Locator (URL) is a URI that specifies the means of acting upon or obtaining the representation of a resource, i.e. specifying both its primary access mechanism and network location. For example, the URL `http://example.org/wiki/Main_Page` refers to a resource identified as `/wiki/Main_Page`, whose representation is obtainable via the Hypertext Transfer Protocol (*http*;) from a network host whose domain name is `example.org`. (In this case, HTTP usually implies it to be in the form of HTML and related code. In practice, that is not necessarily the case, as HTTP allows specifying arbitrary formats in its header.)

A URN is analogous to a person's name, while a URL is analogous to their street address. In other words, a URN identifies an item and a URL provides a method for finding it.

Technical publications, especially standards produced by the IETF and by the W3C, normally reflect a view outlined in a W3C Recommendation of 30 July 2001, which acknowledges the precedence of the term URI rather than endorsing any formal subdivision into URL and URN.

URL is a useful but informal concept: a URL is a type of URI that identifies a resource via a representation of its primary access mechanism (e.g., its network "location"), rather than by some other attributes it may have.^[16]

As such, a URL is simply a URI that happens to point to a resource over a network.^{[a][13]} However, in non-technical contexts and in software for the World Wide Web, the term "URL" remains widely used. Additionally, the term "web address" (which has no formal definition) often occurs in non-technical publications as a synonym for a URI that uses the *http* or *https* schemes. Such assumptions can lead to confusion, for example, in the case of XML namespaces that have a visual similarity to resolvable URIs.

Specifications produced by the WHATWG prefer *URL* over *URI*, and so newer HTML5 APIs use

URL over URI.^[17]

Standardize on the term URL. URI and IRI [Internationalized Resource Identifier] are just confusing. In practice a single algorithm is used for both so keeping them distinct is not helping anyone. URL also easily wins the search result popularity contest.^[18]

While most URI schemes were originally designed to be used with a particular protocol, and often have the same name, they are semantically different from protocols. For example, the scheme *http* is generally used for interacting with web resources using HTTP, but the scheme *file* has no protocol.

Syntax

A URI has a scheme that refers to a specification for assigning identifiers within that scheme. As such, the URI syntax is a federated and extensible naming system wherein each scheme's specification may further restrict the syntax and semantics of identifiers using that scheme. The URI generic syntax is a superset of the syntax of all URI schemes. It was first defined in RFC 2396 (<https://www.rfc-editor.org/rfc/rfc2396>), published in August 1998,^[9] and finalized in RFC 3986 (<https://www.rfc-editor.org/rfc/rfc3986>), published in January 2005.^[19]

A URI is composed from an allowed set of ASCII characters consisting of reserved characters (gen-delims: :, /, ?, #, [,], and @; sub-delims: !, \$, &, ', (,), *, +, ,, ;, and =),^{[1]:13–14} unreserved characters (uppercase and lowercase letters, decimal digits, −, ., _, and ~),^{[1]:13–14} and the character %.^{[1]:12} Syntax components and subcomponents are separated by *delimiters* from the reserved characters (only from generic reserved characters for components) and define *identifying data* represented as unreserved characters, reserved characters that do not act as delimiters in the component and subcomponent respectively,^{[1]:§2} and percent-encodings when the corresponding character is outside the allowed set or is being used as a delimiter of, or within, the component. A percent-encoding of an identifying data octet is a sequence of three characters, consisting of the character % followed by the two hexadecimal digits representing that octet's numeric value.^{[1]:§2.1}

The URI generic syntax consists of five *components* organized hierarchically in order of decreasing significance from left to right:^{[1]:§3}

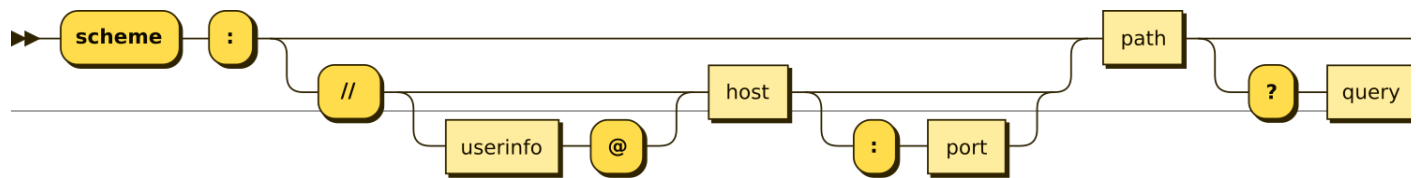
```
URI = scheme ":" ["/" authority] path ["?" query] ["#" fragment]
```

A component is *undefined* if it has an associated delimiter and the delimiter does not appear in the URI; the scheme and path components are always defined.^{[1]:§5.2.1} A component is *empty* if it has no characters; the scheme component is always non-empty.^{[1]:§3}

The authority component consists of *subcomponents*:

```
authority = [userinfo "@"] host [":" port]
```

This is represented in a syntax diagram as:



The URI comprises:

- A non-empty **scheme** component followed by a colon (:), consisting of a sequence of characters beginning with a letter and followed by any combination of letters, digits, plus (+), period (.), or hyphen (-). Although schemes are case-insensitive, the canonical form is lowercase and documents that specify schemes must do so with lowercase letters. Examples of popular schemes include `http`, `https`, `ftp`, `mailto`, `file`, `data` and `irc`. URI schemes should be registered with the Internet Assigned Numbers Authority (IANA), although non-registered schemes are used in practice.^[20]
- An optional **authority** component preceded by two slashes (//), comprising:
 - An optional **userinfo** subcomponent followed by an at symbol (@), that may consist of a user name and an optional password preceded by a colon (:). Use of the format `username:password` in the userinfo subcomponent is deprecated for security reasons. Applications should not render as clear text any data after the first colon (:) found within a userinfo subcomponent unless the data after the colon is the empty string (indicating no password).
 - A **host** subcomponent, consisting of either a registered name (including but not limited to a hostname) or an IP address. IPv4 addresses must be in dot-decimal notation, and IPv6 addresses must be enclosed in brackets ([]).^{[1]:§3.2.2[b]}
 - An optional **port** subcomponent preceded by a colon (:), consisting of decimal digits.
- A **path** component, consisting of a sequence of path segments separated by a slash (/). A path is always defined for a URI, though the defined path may be empty (zero length). A segment may also be empty, resulting in two consecutive slashes (//) in the path component. A path component may resemble or map exactly to a file system path but does not always imply a relation to one. If an authority component is defined, then the path component must either be empty or begin with a slash (/). If an authority component is undefined, then the path cannot begin with an empty segment—that is, with two slashes (//)—since the following characters would be interpreted as an authority component.^{[9]:§3.3}

By convention, in **http** and **https** URIs, the last part of a *path* is named **pathinfo** and it is optional. It is composed by zero or more path segments that do not refer to an existing physical resource name (e.g. a file, an internal module program or an executable program) but to a logical part (e.g. a command or a qualifier part) that has to be passed separately to the first part of the path that identifies an executable module or program managed by a web server; this is often used to select dynamic content (a document, etc.) or to tailor it as requested (see also: CGI and PATH_INFO, etc.).

Example:

URI: "`http://www.example.com/questions/3456/my-document`"

where: `/questions` is the first part of the *path* (an executable module or program) and `/3456/my-document` is the second part of the *path* named *pathinfo*, which is passed to the executable module or program named `/questions` to select the requested document.

An **http** or **https** URI containing a *pathinfo* part without a query part may also be referred to as a 'clean URL,' whose last part may be a 'slug.'

- An optional **query** component preceded by a question mark (?), consisting of a query string of

non-hierarchical data. Its syntax is not well defined, but by convention is most often a sequence of attribute–value pairs separated by a delimiter.

- An optional **fragment** component preceded by a hash (#). The fragment contains a fragment identifier providing direction to a secondary resource, such as a section heading in an article identified by the remainder of the URI. When the primary resource is an HTML document, the fragment is often an id attribute of a specific element, and web browsers will scroll this element into view.

Query delimiter	Example
Ampersand (&)	key1=value1&key2=value2
Semicolon (;) ^[c]	key1=value1;key2=value2

The scheme- or implementation-specific reserved character + may be used in the scheme, userinfo, host, path, query, and fragment, and the scheme- or implementation-specific reserved characters !, \$, &, ', (,), *, ,, ;, and = may be used in the userinfo, host, path, query, and fragment. Additionally, the generic reserved character : may be used in the userinfo, path, query and fragment, the generic reserved characters @ and / may be used in the path, query and fragment, and the generic reserved character ? may be used in the query and fragment.^{[1]:§A}

Example URIs

The following figure displays example URIs and their component parts.

DOIs (digital object identifiers) fit within the Handle System and fit within the URI system, as facilitated by appropriate syntax.

URI references

A *URI reference* is either a URI or a *relative reference* when it does not begin with a scheme component followed by a colon (:).^{[1]:§4.1} A path segment that contains a colon character (e.g., `foo:bar`) cannot be used as the first path segment of a relative reference if its path component does not begin with a slash (/), as it would be mistaken for a scheme component. Such a path segment must be preceded by a dot path segment (e.g., `./foo:bar`).^{[1]:§4.2}

Web document markup languages frequently use URI references to point to other resources, such as external documents or specific portions of the same logical document:^{[1]:§4.4}

- in HTML, the value of the `src` attribute of the `img` element provides a URI reference, as does the value of the `href` attribute of the `a` or `link` element;
- in XML, the system identifier appearing after the `SYSTEM` keyword in a DTD is a fragmentless URI reference;
- in XSLT, the value of the `href` attribute of the `xsl:import` element/instruction is a URI reference; likewise the first argument to the `document()` function.

[illegible]

```
ce.txt
/path/resource.txt
path/resource.txt
../resource.txt
./resource.txt
resource.txt
#fragment
```

Resolution

Resolving a URI reference against a *base URI* results in a *target URI*. This implies that the base URI exists and is an *absolute URI* (a URI with no fragment component). The base URI can be obtained, in order of precedence, from:^{[1]:§5.1}

- the reference URI itself if it is a URI;
- the content of the representation;
- the entity encapsulating the representation;
- the URI used for the actual retrieval of the representation;
- the context of the application.

Within a representation with a well defined base URI of

```
http://a/b/c/d;p?q
```

a relative reference is resolved to its target URI as follows:^{[1]:§5.4}

```
"g:h"    -> "g:h"
"g"       -> "http://a/b/c/g"
"./g"     -> "http://a/b/c/g"
"g/"      -> "http://a/b/c/g/"
"/g"      -> "http://a/g"
"/g/"     -> "http://g"
"?y"      -> "http://a/b/c/d;p?q?y"
"g?y"     -> "http://a/b/c/g?y"
"#s"      -> "http://a/b/c/d;p?q#s"
"g#s"     -> "http://a/b/c/g#s"
"g?y#s"   -> "http://a/b/c/g?y#s"
";x"      -> "http://a/b/c/;x"
"g;x"     -> "http://a/b/c/g;x"
"g;x?y#s" -> "http://a/b/c/g;x?y#s"
""        -> "http://a/b/c/d;p?q"
"."       -> "http://a/b/c/"
"./"      -> "http://a/b/c/"
".."     -> "http://a/b/"
"./."    -> "http://a/b/"
"./g"    -> "http://a/b/g"
"../.."  -> "http://a/"
"../../" -> "http://a/"
"../..g" -> "http://a/g"
```

URL munging

URL munging is a technique by which a command is appended to a URL, usually at the end, after a "?" token. It is commonly used in WebDAV as a mechanism of adding functionality to HTTP. In a versioning system, for example, to add a "checkout" command to a URL, it is written as `http://`

`editing.com/resource/file.php?command=checkout`. It has the advantage of both being easy for [CGI parsers](#) and also acts as an intermediary between HTTP and underlying resource, in this case.^[24]

Relation to XML namespaces

In [XML](#), a [namespace](#) is an abstract domain to which a collection of element and attribute names can be assigned. The namespace name is a character string which must adhere to the generic URI syntax.^[25] However, the name is generally not considered to be a URI,^[26] because the URI specification bases the decision not only on lexical components, but also on their intended use. A namespace name does not necessarily imply any of the semantics of URI schemes; for example, a namespace name beginning with *http*: may have no connotation to the use of the [HTTP](#).

Originally, the namespace name could match the syntax of any non-empty URI reference, but the use of relative URI references was deprecated by the W3C.^[27] A separate W3C specification for namespaces in XML 1.1 permits [Internationalized Resource Identifier](#) (IRI) references to serve as the basis for namespace names in addition to URI references.^[28]

See also

- [CURIE](#)
- [Linked data](#)
- [Extensible Resource Identifier](#)
- [Internationalized Resource Identifier \(IRI\)](#)
- [Internet resource locator](#)
- [Persistent uniform resource locator](#)
- [Uniform Naming Convention](#)
- [Resource Directory Description Language](#)
- [Universally unique identifier](#)
- [List of URI schemes](#)
- [Resource Description Framework](#)

Notes

- A report published in 2002 by a joint W3C/IETF working group aimed to normalize the divergent views held within the IETF and W3C over the relationship between the various 'UR*' terms and standards. While not published as a full standard by either organization, it has become the basis for the above common understanding and has informed many standards since then.
- For URIs relating to resources on the World Wide Web, some web browsers allow `.0` portions of dot-decimal notation to be dropped or raw integer IP addresses to be used.^[21]
- Historic RFC 1866 (<https://www.rfc-editor.org/rfc/rfc1866>) (obsoleted by RFC 2854 (<https://www.rfc-editor.org/rfc/rfc2854>)^[22]) encourages CGI authors to support `';` in addition to `'&'`.^[23]§8.2.1

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Further reading

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External links

- URI Schemes (<https://www.iana.org/assignments/uri-schemes/uri-schemes.xhtml>) – IANA-maintained registry of URI Schemes
- URI schemes on the W3C wiki (<https://www.w3.org/wiki/UriSchemes>)
- Architecture of the World Wide Web, Volume One, §2: Identification (<https://www.w3.org/TR/webarch/#identification>) – by W3C
- W3C URI Clarification (<https://www.w3.org/TR/uri-clarification/>)

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